Lesson 10: Random Variables and Distributions

AI/ML Mathematics

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1 Motivation and Overview

A reinforcement learning agent interacting with an environment receives numerical rewards at each time step. The reward after pulling arm k of a multi-armed bandit might be $+1$ (success) or 0 (failure); the total number of successes in n pulls could be any integer from 0 to n; the time until the first success is a positive integer; and the noise added to a policy gradient estimate is often modeled as a continuous bell-shaped quantity. Each of these quantities is a function that assigns a real number to every possible experimental outcome. How should we formalize such numerical summaries of random experiments? What tools let us compute the probability that a random quantity falls in a given range, and what standard families of distributions arise repeatedly in practice?

The tools developed in this lesson will let us formalize random variables as measurable functions from a sample space to the real line, describe their distributions through probability mass functions, probability density functions, and cumulative distribution functions, and catalog the discrete and continuous distributions most important for reinforcement learning and machine learning.

Why Random Variables and Distributions Matter for RL/ML

- **Bandit rewards**: Each arm of a multi-armed bandit produces Bernoulli-distributed rewards; the agent must estimate the mean of each arm's distribution.
- **Event counting in RL**: The number of times a rare event occurs in n episodes is approximately Poisson-distributed, relevant for safety monitoring and anomaly detection.
- **Policy gradient noise**: The REINFORCE estimator adds Gaussian noise to gradient estimates; understanding the normal distribution is essential for variance analysis and confidence intervals.

Prerequisites. Probability spaces, Kolmogorov axioms, conditional probability, Bayes' theorem, and independence (Lesson 9). Familiarity with sequences, series, and basic integral calculus (Lessons 2–4).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* random variables, probability mass functions, probability density functions, and cumulative distribution functions.
- *Derive* the PMF, expectation, and variance of the Bernoulli, Binomial, Geometric, and Poisson distributions.
- *Prove* the Poisson limit theorem (Poisson approximation to the Binomial).
- *Compute* the PDF, CDF, expectation, and variance of the Uniform, Exponential, and Normal distributions.
- *Prove* the Gaussian integral formula using polar coordinates.
- *Explain* why $\mathbb{P}(X = x) = 0$ for continuous random variables and why density values are not probabilities.

2 References

- Ross [2019, Chapters 3–5] – Random variables, discrete and continuous distributions with many worked examples.

- Blitzstein and Hwang [2019, Chapters 3–5] – Random variables and their distributions with a storytelling approach and extensive exercises.
- Feller [1968, Chapters 2–4] – Classical treatment of discrete distributions and generating functions.
- Grimmett and Stirzaker [2020, Chapter 2] – Random variables, distribution functions, and density functions in a rigorous framework.
- Bertsekas and Tsitsiklis [2008, Chapter 2] – Discrete and continuous random variables with engineering applications.

3 Random Variables and Distribution Functions

We begin by formalizing the notion of a numerical quantity determined by the outcome of a random experiment.

Definition 3.1 (Random Variable)**.** Let $(\Omega, \mathbb{P})$ be a probability space. A *random variable* is a function $X\colon \Omega \to \mathbb{R}$ that assigns a real number $X(\omega)$ to each outcome $\omega \in \Omega$.

Remark 3.2. In this lesson we work with discrete or "well-behaved" continuous sample spaces where every function $X\colon \Omega \to \mathbb{R}$ is automatically measurable. For uncountable sample spaces equipped with a σ-algebra $\mathcal{F}$, one must additionally require that $\{\omega : X(\omega) \le x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$; this refinement is treated in Phase 01 (Lesson 1, measurable functions).

Definition 3.3 (Cumulative Distribution Function)**.** The *cumulative distribution function* (CDF) of a random variable X is the function $F_X\colon \mathbb{R} \to [0,1]$ defined by

$$F_X(x) = \mathbb{P}(X \le x) \quad \text{for all } x \in \mathbb{R}.$$

Proposition 3.4 (Properties of the CDF)**.** *For any random variable X, the CDF F_X satisfies:*
(i) F_X is non-decreasing: if $x \le y$, then $F_X(x) \le F_X(y)$.
(ii) $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$.
(iii) F_X is right-continuous: $\lim_{h\to 0^+} F_X(x+h) = F_X(x)$ for all $x \in \mathbb{R}$.

Proof. **(i)** If $x \le y$, then $\{X \le x\} \subseteq \{X \le y\}$, so by monotonicity of $\mathbb{P}$, $F_X(x) = \mathbb{P}(X \le x) \le \mathbb{P}(X \le y) = F_X(y)$.

(ii) The events $A_n = \{X \le -n\}$ satisfy $A_1 \supseteq A_2 \supseteq \cdots$ and $\bigcap_{n=1}^{\infty} A_n = \emptyset$. By continuity from above (Lesson 9, Theorem 3.5), $\lim_{n\to\infty} F_X(-n) = \mathbb{P}(\emptyset) = 0$. Since F_X is non-decreasing, $\lim_{x\to-\infty} F_X(x) = 0$. Similarly, $B_n = \{X \le n\}$ gives an increasing sequence with $\bigcup B_n = \Omega$, so $\lim_{x\to+\infty} F_X(x) = 1$ by continuity from below.

(iii) Let $h_n \downarrow 0$. Then $\{X \le x + h_n\} \downarrow \{X \le x\}$ (since any ω with $X(\omega) \le x$ is in every $\{X \le x + h_n\}$, and any ω with $X(\omega) > x$ satisfies $X(\omega) > x + h_n$ for n large enough). By continuity from above, $\lim_n F_X(x + h_n) = F_X(x)$. □

Example 3.5 (CDF of a Coin Flip)**.** Toss a fair coin. Define $X = 1$ if heads, $X = 0$ if tails. Then

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1/2 & \text{if } 0 \le x < 1, \\ 1 & \text{if } x \ge 1. \end{cases}$$

The CDF has jumps of size $1/2$ at $x = 0$ and $x = 1$.

We now distinguish two fundamental types of random variables: discrete and continuous.

4 Discrete Random Variables

Definition 4.1 (Discrete Random Variable and PMF)**.** A random variable X is *discrete* if it takes values in a countable set $\{x_1, x_2, \ldots\}$. Its *probability mass function* (PMF) is

$$p_X(x) = \mathbb{P}(X = x) \quad \text{for each } x \in \mathbb{R}.$$

The PMF satisfies $p_X(x) \geq 0$ for all x, $p_X(x) = 0$ for $x \notin \{x_1, x_2, \ldots\}$, and $\sum_i p_X(x_i) = 1$.

Definition 4.2 (Expectation of a Discrete Random Variable)**.** Let X be a discrete random variable with PMF p_X. The *expectation* (or *mean*) of X is

$$\mathbb{E}[X] = \sum_x x\, p_X(x),$$

provided the sum converges absolutely, i.e., $\sum_x |x|\, p_X(x) < \infty$.

Definition 4.3 (Variance of a Discrete Random Variable)**.** The *variance* of a discrete random variable X with $\mathbb{E}[X^2] < \infty$ is

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \sum_x (x - \mathbb{E}[X])^2\, p_X(x).$$

The *standard deviation* is $\sigma_X = \sqrt{\mathrm{Var}(X)}$.

Proposition 4.4 (Computational Formula for Variance)**.** *For any random variable X with finite second moment,*

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Proof. Let $\mu = \mathbb{E}[X]$. Then

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2.\end{aligned}$$ □

We now catalog the most important discrete distributions.

4.1 Bernoulli Distribution

Definition 4.5 (Bernoulli Distribution)**.** A random variable X has a *Bernoulli distribution* with parameter $p \in [0, 1]$, written $X \sim \mathrm{Bernoulli}(p)$, if X takes values in $\{0, 1\}$ with PMF

$$p_X(k) = p^k (1-p)^{1-k}, \quad k \in \{0, 1\}.$$

That is, $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$.

Proposition 4.6 (Bernoulli Moments)**.** *If $X \sim \mathrm{Bernoulli}(p)$, then $\mathbb{E}[X] = p$ and $\mathrm{Var}(X) = p(1-p)$.*

Proof. Since $X \in \{0, 1\}$, we have $X^2 = X$, so $\mathbb{E}[X] = 0 \cdot (1-p) + 1 \cdot p = p$ and $\mathbb{E}[X^2] = \mathbb{E}[X] = p$. By Proposition 4.4, $\mathrm{Var}(X) = p - p^2 = p(1-p)$. □

Bernoulli Rewards in Multi-Armed Bandits

In a K-armed Bernoulli bandit, pulling arm k yields a reward $R_k \sim \mathrm{Bernoulli}(p_k)$. The agent's goal is to maximize the total expected reward $\sum_{t=1}^{T} \mathbb{E}[R_{A_t}]$ where A_t is the arm selected at time t. The regret analysis of algorithms like UCB and Thompson sampling

relies on the Bernoulli mean p_k and variance $p_k(1-p_k)$ to quantify the difficulty of distinguishing near-optimal arms.

Example 4.7 (Bernoulli Variance)**.** The variance $p(1-p)$ is maximized at $p = 1/2$ (most uncertain case) and equals zero at $p = 0$ or $p = 1$ (deterministic outcomes). This is why a fair coin flip is the "hardest" Bernoulli trial to predict: it has the largest spread.

4.2 Binomial Distribution

Definition 4.8 (Binomial Distribution)**.** A random variable X has a *Binomial distribution* with parameters $n \in \mathbb{N}$ and $p \in [0,1]$, written $X \sim \mathrm{Bin}(n,p)$, if

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Proposition 4.9 (Binomial PMF is Valid)**.** *The Binomial PMF sums to* 1.

Proof. By the binomial theorem,

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = \big(p + (1-p)\big)^n = 1^n = 1. \qquad \square$$

Theorem 4.10 (Binomial Moments via Indicator Decomposition)**.** *If* $X \sim \mathrm{Bin}(n,p)$*, then* $\mathbb{E}[X] = np$ *and* $\mathrm{Var}(X) = np(1-p)$*.*

Proof. Write $X = X_1 + X_2 + \cdots + X_n$ where each $X_i \sim \mathrm{Bernoulli}(p)$ independently represents the outcome of the i-th trial. Since expectation is additive (regardless of independence; see Lesson 11),

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np.$$

For the variance, independence gives $\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = n \cdot p(1-p)$. (The variance of a sum of independent random variables equals the sum of variances; this is proved in Lesson 11, Theorem 4.3.) $\square$

Example 4.11 (Quality Control)**.** A production line has a 2% defect rate. In a batch of $n = 100$ items, the number of defectives $X \sim \mathrm{Bin}(100, 0.02)$ has $\mathbb{E}[X] = 2$ and $\mathrm{Var}(X) = 100 \cdot 0.02 \cdot 0.98 = 1.96$, so $\sigma_X \approx 1.40$. Most batches will have between 0 and 5 defects.

4.3 Geometric Distribution

Definition 4.12 (Geometric Distribution)**.** A random variable X has a *Geometric distribution* with parameter $p \in (0,1]$, written $X \sim \mathrm{Geom}(p)$, if X counts the number of the trial on which the first success occurs in a sequence of independent Bernoulli(p) trials. Its PMF is

$$p_X(k) = (1-p)^{k-1} p, \quad k = 1, 2, 3, \ldots$$

Proposition 4.13 (Geometric PMF is Valid)**.** *The Geometric PMF sums to* 1.

Proof. Let $q = 1 - p$. Then

$$\sum_{k=1}^{\infty} q^{k-1} p = p \sum_{j=0}^{\infty} q^j = p \cdot \frac{1}{1-q} = \frac{p}{p} = 1,$$

where the geometric series converges since $0 \le q < 1$. $\square$

Proposition 4.14 (Geometric Moments)**.** *If* $X \sim \text{Geom}(p)$*, then* $\mathbb{E}[X] = 1/p$ *and* $\text{Var}(X) = (1-p)/p^2$*.*

Proof. Let $q = 1-p$. For the expectation, we differentiate the geometric series. Recall $\sum_{k=0}^{\infty} q^k = 1/(1-q)$. Differentiating both sides with respect to q: $\sum_{k=1}^{\infty} kq^{k-1} = 1/(1-q)^2 = 1/p^2$. Therefore

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k\, q^{k-1} p = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

For the second moment, differentiate $\sum_{k=1}^{\infty} kq^{k-1} = 1/p^2$ with respect to q: $\sum_{k=2}^{\infty} k(k-1)q^{k-2} = 2/p^3$. Then

$$\mathbb{E}[X(X-1)] = \sum_{k=1}^{\infty} k(k-1)q^{k-1}p = p \cdot q \cdot \frac{2}{p^3} = \frac{2q}{p^2}.$$

So $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = 2q/p^2 + 1/p = (2q+p)/p^2 = (q+1)/p^2$. Using Proposition 4.4,

$$\text{Var}(X) = \frac{q+1}{p^2} - \frac{1}{p^2} = \frac{q}{p^2} = \frac{1-p}{p^2}.$$ □

Example 4.15 (Expected Waiting Time)**.** A server processes requests, each succeeding independently with probability $p = 0.3$. The expected number of attempts until the first success is $\mathbb{E}[X] = 1/0.3 \approx 3.33$, with standard deviation $\sigma_X = \sqrt{0.7/0.09} \approx 2.79$.

4.4 Poisson Distribution

Definition 4.16 (Poisson Distribution)**.** A random variable X has a *Poisson distribution* with parameter $\lambda > 0$, written $X \sim \text{Poisson}(\lambda)$, if

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

Proposition 4.17 (Poisson PMF is Valid)**.** *The Poisson PMF sums to* 1*.*

Proof. Using the Taylor series $e^{\lambda} = \sum_{k=0}^{\infty} \lambda^k/k!$,

$$\sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot e^{\lambda} = 1.$$ □

Proposition 4.18 (Poisson Moments)**.** *If* $X \sim \text{Poisson}(\lambda)$*, then* $\mathbb{E}[X] = \lambda$ *and* $\text{Var}(X) = \lambda$*.*

Proof. For the expectation:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda.$$

For $\mathbb{E}[X(X-1)]$:

$$\mathbb{E}[X(X-1)] = \sum_{k=2}^{\infty} \frac{k(k-1)\lambda^k e^{-\lambda}}{k!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda^2.$$

Therefore $\mathbb{E}[X^2] = \lambda^2 + \lambda$, and $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$. □

Poisson Limit Theorem

The Poisson distribution arises as the limit of the Binomial distribution when $n \to \infty$ and $p \to 0$ such that $np \to \lambda$. This justifies using the Poisson as an approximation for counting rare events in many independent trials.

Theorem 4.19 (Poisson Limit Theorem)**.** *Let $X_n \sim \mathrm{Bin}(n, p_n)$ where $p_n = \lambda/n$ for a fixed $\lambda > 0$. Then for every $k = 0, 1, 2, \ldots$,*

$$\lim_{n\to\infty} \mathbb{P}(X_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

Proof. We have

$$\mathbb{P}(X_n = k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

Write $\binom{n}{k} = n(n-1)\cdots(n-k+1)/k!$ and factor:

$$\begin{aligned}\mathbb{P}(X_n = k) &= \frac{n(n-1)\cdots(n-k+1)}{k!} \cdot \frac{\lambda^k}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-k+1}{n}}_{\to 1 \text{ as } n\to\infty} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\to 1}.\end{aligned}$$

The first underbraced factor is a product of k terms each tending to 1. The second factor converges to $e^{-\lambda}$ by the well-known limit $\lim_{n\to\infty}(1 - \lambda/n)^n = e^{-\lambda}$. The third factor tends to 1 since k is fixed and $(1 - \lambda/n) \to 1$. Combining: $\lim_{n\to\infty} \mathbb{P}(X_n = k) = \lambda^k e^{-\lambda}/k!$. □

Poisson Processes in RL Event Counting

In safety-critical RL applications, an agent may need to monitor the occurrence of rare failure events across many episodes. If each episode independently has a small failure probability p, the total failure count across n episodes is approximately $\mathrm{Poisson}(np)$ by the Poisson limit theorem. This approximation simplifies the computation of tail probabilities for alarm thresholds and risk-sensitive objectives.

Example 4.20 (Poisson Approximation)**.** A website receives $n = 10{,}000$ visitors per hour, each clicking a specific ad with probability $p = 0.0003$. The exact distribution of clicks is $\mathrm{Bin}(10000, 0.0003)$, but the Poisson approximation gives $\lambda = np = 3$. Then $\mathbb{P}(X = 0) \approx e^{-3} \approx 0.050$ and $\mathbb{P}(X \le 2) \approx e^{-3}(1 + 3 + 9/2) \approx 0.423$.

Having covered the key discrete distributions, we now turn to continuous random variables.

5 Continuous Random Variables

Definition 5.1 (Continuous Random Variable and PDF)**.** A random variable X is *continuous* if there exists a non-negative function $f_X \colon \mathbb{R} \to [0, \infty)$, called the *probability density function* (PDF), such that for every interval $[a, b]$,

$$\mathbb{P}(a \le X \le b) = \int_a^b f_X(x)\, dx.$$

The PDF satisfies $f_X(x) \ge 0$ for all x and $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$.

Proposition 5.2 (CDF-PDF Relationship)**.** *If X is a continuous random variable with PDF f_X, then*

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt,$$

and at every point where f_X is continuous, $f_X(x) = F_X'(x)$.

Proof. By definition, $F_X(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f_X(t)\,dt$. By the Fundamental Theorem of Calculus, at points of continuity of f_X, the derivative of this integral equals $f_X(x)$. □

$\mathbb{P}(X = x) = 0$ for Continuous Random Variables

For a continuous random variable, $\mathbb{P}(X = x) = 0$ for every $x \in \mathbb{R}$. This is because $\mathbb{P}(X = x) = \mathbb{P}(x \le X \le x) = \int_x^x f_X(t)\,dt = 0$. In particular, $f_X(x)$ is *not* the probability that X equals x; rather, $f_X(x)\,dx$ represents an infinitesimal probability. Density values can exceed 1 (e.g., the PDF of Uniform$(0, 1/2)$ equals 2 on $(0, 1/2)$).

Definition 5.3 (Expectation and Variance of Continuous Random Variables)**.** If X is continuous with PDF f_X, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\,dx$$

provided the integral converges absolutely, and

$$\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2\, f_X(x)\,dx = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

5.1 Uniform Distribution

Definition 5.4 (Continuous Uniform Distribution)**.** A random variable X has a *Uniform distribution* on $[a, b]$ with $a < b$, written $X \sim \mathrm{Uniform}(a, b)$, if its PDF is

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \le x \le b, \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 5.5 (Uniform Moments)**.** *If $X \sim \mathrm{Uniform}(a, b)$, then $\mathbb{E}[X] = (a + b)/2$ and $\mathrm{Var}(X) = (b - a)^2/12$.*

Proof.

$$\mathbb{E}[X] = \int_a^b \frac{x}{b-a}\,dx = \frac{1}{b-a} \cdot \frac{x^2}{2}\bigg|_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

For the second moment,

$$\mathbb{E}[X^2] = \int_a^b \frac{x^2}{b-a}\,dx = \frac{1}{b-a} \cdot \frac{x^3}{3}\bigg|_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}.$$

Therefore

$$\begin{aligned} \mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 &= \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} \\ &= \frac{4(a^2 + ab + b^2) - 3(a^2 + 2ab + b^2)}{12} = \frac{a^2 - 2ab + b^2}{12} = \frac{(b-a)^2}{12}. \end{aligned}$$ □

Example 5.6 (Uniform Random Number Generator)**.** For $X \sim \mathrm{Uniform}(0, 1)$, we have $\mathbb{E}[X] = 1/2$ and $\mathrm{Var}(X) = 1/12 \approx 0.083$. The CDF is $F_X(x) = x$ for $x \in [0, 1]$, which is the basis for the inverse transform method: if $U \sim \mathrm{Uniform}(0, 1)$ and F is any CDF, then $X = F^{-1}(U)$ has CDF F.

5.2 Exponential Distribution

Definition 5.7 (Exponential Distribution)**.** A random variable X has an *Exponential distribution* with rate parameter $\lambda > 0$, written $X \sim \mathrm{Exp}(\lambda)$, if its PDF is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Proposition 5.8 (Exponential Moments)**.** *If $X \sim \mathrm{Exp}(\lambda)$, then $\mathbb{E}[X] = 1/\lambda$ and $\mathrm{Var}(X) = 1/\lambda^2$.*

Proof. Using integration by parts with $u = x$, $dv = \lambda e^{-\lambda x} dx$:

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x}\, dx = \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\, dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.$$

For $\mathbb{E}[X^2]$, integrate by parts twice:

$$\mathbb{E}[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x}\, dx = \frac{2}{\lambda^2}.$$

(Alternatively, $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X]$; using the substitution $j = k - 2$ in the factorial moment computation gives the same result.) Therefore $\mathrm{Var}(X) = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$. □

Theorem 5.9 (Memoryless Property of the Exponential Distribution)**.** *Let $X \sim \mathrm{Exp}(\lambda)$. Then for all $s, t \geq 0$,*

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Moreover, the Exponential distribution is the unique continuous distribution with the memoryless property.

Proof. The CDF is $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$, so $\mathbb{P}(X > x) = e^{-\lambda x}$. By the definition of conditional probability,

$$\mathbb{P}(X > s + t \mid X > s) = \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(X > t).$$

For uniqueness, suppose X is a continuous non-negative random variable satisfying $\mathbb{P}(X > s + t) = \mathbb{P}(X > s)\mathbb{P}(X > t)$ for all $s, t \geq 0$. Let $g(t) = \mathbb{P}(X > t)$. Then $g(s + t) = g(s)g(t)$, $g(0) = 1$, and g is non-increasing and right-continuous (since $F_X = 1 - g$ is a CDF). The only non-increasing solutions to Cauchy's functional equation $g(s + t) = g(s)g(t)$ with $g(0) = 1$ that are measurable (guaranteed by monotonicity) are $g(t) = e^{-\lambda t}$ for some $\lambda \geq 0$. Since $g(t) \to 0$ as $t \to \infty$ (because X is a proper random variable), we need $\lambda > 0$. This gives $X \sim \mathrm{Exp}(\lambda)$. □

Example 5.10 (Exponential Waiting Time)**.** A light bulb has an exponentially distributed lifetime with mean $1/\lambda = 1000$ hours. Given that it has survived 500 hours, the memoryless property says the conditional distribution of the remaining lifetime is still $\mathrm{Exp}(\lambda)$ with mean 1000 hours. The bulb's past survival provides no information about its future.

5.3 Normal (Gaussian) Distribution

Definition 5.11 (Normal Distribution)**.** A random variable X has a *Normal* (or *Gaussian*) distribution with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

The case $\mu = 0$, $\sigma^2 = 1$ is the *standard normal* distribution.

To verify that this is a valid PDF, we must show that the integral over $\mathbb{R}$ equals 1. It suffices to evaluate the integral for the standard normal.

The Gaussian Integral

The integral $\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$ is the foundational identity that normalizes the Gaussian density. It cannot be evaluated by elementary antiderivatives; the proof uses a change to polar coordinates.

Theorem 5.12 (Gaussian Integral via Polar Coordinates)**.**

$$I = \int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}.$$

Proof. Since $e^{-x^2/2} > 0$, the integral I is positive (and converges because $e^{-x^2/2}$ decays faster than any polynomial). We compute I^2 instead:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\,dx\,dy.$$

Switch to polar coordinates $x = r\cos\theta$, $y = r\sin\theta$ with Jacobian r:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\,r\,dr\,d\theta = 2\pi\int_0^{\infty} r\,e^{-r^2/2}\,dr.$$

Substitute $u = r^2/2$, $du = r\,dr$:

$$\int_0^{\infty} r\,e^{-r^2/2}\,dr = \int_0^{\infty} e^{-u}\,du = \left[-e^{-u}\right]_0^{\infty} = 1.$$

Therefore $I^2 = 2\pi$, and since $I > 0$, we conclude $I = \sqrt{2\pi}$. □

Corollary 5.13. *The standard normal PDF $f(x) = (2\pi)^{-1/2}e^{-x^2/2}$ integrates to 1 over $\mathbb{R}$. For general $\mathcal{N}(\mu,\sigma^2)$, the substitution $z = (x-\mu)/\sigma$ with $dx = \sigma\,dz$ gives*

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz = 1.$$

Theorem 5.14 (Normal Moments via Standardization)**.** *If $X \sim \mathcal{N}(\mu,\sigma^2)$, then $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$.*

Proof. Standard normal first. Let $Z \sim \mathcal{N}(0,1)$. Since $z\,e^{-z^2/2}$ is an odd function and $e^{-z^2/2}$ decays rapidly,

$$\mathbb{E}[Z] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} z\,e^{-z^2/2}\,dz = 0.$$

For the second moment, integrate by parts with $u = z$, $dv = z\,e^{-z^2/2}\,dz$ (so $v = -e^{-z^2/2}$):

$$\begin{aligned}\mathbb{E}[Z^2] &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\,dz = \frac{1}{\sqrt{2\pi}}\left(\left[-z\,e^{-z^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-z^2/2}\,dz\right)\\ &= \frac{1}{\sqrt{2\pi}}\left(0 + \sqrt{2\pi}\right) = 1.\end{aligned}$$

Thus $\mathrm{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 = 1 - 0 = 1$.

General case. Write $X = \mu + \sigma Z$ where $Z \sim \mathcal{N}(0,1)$. Then $\mathbb{E}[X] = \mu + \sigma\mathbb{E}[Z] = \mu$ and $\mathrm{Var}(X) = \sigma^2\,\mathrm{Var}(Z) = \sigma^2$ (using the affine transformation rule $\mathrm{Var}(aZ+b) = a^2\,\mathrm{Var}(Z)$, proved in Lesson 11). □

Gaussian Noise in Policy Gradient Methods

In continuous-action policy gradient methods such as REINFORCE, the policy $\pi_\theta(a \mid s)$ is typically a Gaussian $\mathcal{N}(\mu_\theta(s), \sigma_\theta^2(s))$. The gradient estimate

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \nabla_\theta \log \pi_\theta(a_i \mid s_i)\, G_i$$

involves the log-density of the normal distribution. Understanding the Gaussian PDF, its mean, and variance is essential for computing $\nabla_\theta \log \pi_\theta$, analyzing estimator variance, and designing variance reduction baselines.

Example 5.15 (Standard Normal Tail Probabilities)**.** For $Z \sim \mathcal{N}(0,1)$, the "68-95-99.7 rule" states

$$\mathbb{P}(|Z| \le 1) \approx 0.683, \quad \mathbb{P}(|Z| \le 2) \approx 0.954, \quad \mathbb{P}(|Z| \le 3) \approx 0.997.$$

This means observations more than 3σ from the mean are extremely rare under Gaussian models, a fact used in outlier detection and confidence interval construction.

6 Exercises

Exercise 6.1 (Binomial Probabilities)**.** $(*)$
A fair coin is tossed $n = 6$ times. Let X be the number of heads.

(a) Compute $\mathbb{P}(X = 3)$.
(b) Compute $\mathbb{P}(X \ge 5)$.
(c) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ using the Binomial formulas.

Exercise 6.2 (Geometric Distribution and the Memoryless Property)**.** $(*)$
Let $X \sim \mathrm{Geom}(p)$ count the trial of the first success.

(a) Prove that the Geometric distribution has the discrete memoryless property: $\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n)$ for all $m, n \ge 0$.
(b) Is the Geometric distribution the unique discrete distribution with this property? Prove or disprove.

Exercise 6.3 (Poisson Approximation Error)**.** $(**)$
Let $X \sim \mathrm{Bin}(n, \lambda/n)$ and $Y \sim \mathrm{Poisson}(\lambda)$ with $\lambda = 2$ and $n = 20$.

(a) Compute $\mathbb{P}(X = 0)$, $\mathbb{P}(X = 1)$, and $\mathbb{P}(X = 2)$ exactly using the Binomial PMF.
(b) Compute the corresponding Poisson approximations $\mathbb{P}(Y = 0)$, $\mathbb{P}(Y = 1)$, $\mathbb{P}(Y = 2)$.
(c) Quantify the approximation error for each value.

Exercise 6.4 (Exponential CDF and Quantiles)**.** $(**)$
Let $X \sim \mathrm{Exp}(\lambda)$.

(a) Derive the CDF $F_X(x)$ from the PDF.
(b) Find the median of X (i.e., the value m such that $F_X(m) = 1/2$).
(c) Show that the median is always less than the mean $1/\lambda$. What does this tell us about the shape of the exponential distribution?

Exercise 6.5 (Gaussian Moments via Differentiation)**.** $(***)$
Define $I(\alpha) = \int_{-\infty}^{\infty} e^{-\alpha x^2}\, dx$ for $\alpha > 0$.

(a) Using the Gaussian integral (Theorem 5.12 with the substitution $u = \sqrt{2\alpha}\, x$), show that $I(\alpha) = \sqrt{\pi/\alpha}$.
(b) Differentiate $I(\alpha)$ with respect to α under the integral sign to derive $\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2}\, dx = \frac{1}{2}\sqrt{\pi/\alpha^3}$.

(c) Use parts (a) and (b) with $\alpha = 1/2$ to re-derive $\mathbb{E}[Z^2] = 1$ for $Z \sim \mathcal{N}(0, 1)$.
Hint: Differentiation under the integral sign is justified by dominated convergence, since $x^2 e^{-\alpha x^2}$ is integrable for $\alpha > 0$.

Exercise 6.6 (Bandit Reward Distributions). $(***)$
A two-armed bandit has arm 1 with reward $R_1 \sim \text{Bernoulli}(0.6)$ and arm 2 with reward $R_2 \sim \mathcal{N}(0.5, 0.25)$.

(a) Compute $\mathbb{E}[R_1]$, $\text{Var}(R_1)$, $\mathbb{E}[R_2]$, and $\text{Var}(R_2)$.
(b) Which arm has the higher expected reward?
(c) Suppose you pull arm 2 and observe $r_2 = 1.3$. Compare the density $f_{R_2}(1.3)$ with the probability $\mathbb{P}(R_1 = 1) = 0.6$. Explain why comparing a density value with a probability is invalid, even though both are numbers associated with the observation.

7 Summary

- A *random variable* $X \colon \Omega \to \mathbb{R}$ assigns a numerical value to each outcome; its distribution is fully characterized by the CDF $F_X(x) = \mathbb{P}(X \le x)$.
- A *discrete* random variable takes countably many values; its distribution is described by the PMF $p_X(k) = \mathbb{P}(X = k)$.
- The *Bernoulli*(p) distribution models a single trial with success probability p; $\mathbb{E}[X] = p$, $\text{Var}(X) = p(1-p)$.
- The *Binomial*(n, p) distribution counts successes in n independent trials; the indicator decomposition $X = \sum X_i$ gives $\mathbb{E}[X] = np$ and $\text{Var}(X) = np(1-p)$.
- The *Geometric*(p) distribution counts the trial of the first success; $\mathbb{E}[X] = 1/p$ and $\text{Var}(X) = (1-p)/p^2$.
- The *Poisson*(λ) distribution models rare event counts; it arises as the limit of $\text{Bin}(n, \lambda/n)$ as $n \to \infty$, and satisfies $\mathbb{E}[X] = \text{Var}(X) = \lambda$.
- A *continuous* random variable has $\mathbb{P}(X = x) = 0$ for every x; its distribution is described by a PDF f_X with $\mathbb{P}(a \le X \le b) = \int_a^b f_X(x)\,dx$.
- The *Exponential*(λ) distribution is the unique continuous memoryless distribution; $\mathbb{E}[X] = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.
- The *Normal*(μ, σ^2) distribution has the bell-shaped density normalized by the Gaussian integral $\int e^{-x^2/2}\,dx = \sqrt{2\pi}$; $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$.

8 Further Reading

- **Lesson 11** develops expectation, variance, covariance, and joint distributions in full generality, including the linearity of expectation and the tower property of conditional expectation.
- **Lesson 12** covers the law of large numbers and the central limit theorem, explaining why sample means converge and why the normal distribution arises so ubiquitously.
- **Phase 01, Lesson 4** (probability foundations) revisits random variables, expectation, and distributions in the measure-theoretic framework using Lebesgue integration.

- For a comprehensive catalog of over one hundred univariate distributions and their relationships, see Blitzstein and Hwang [2019, Appendix A].
- The Poisson distribution was introduced by Siméon Denis Poisson in 1837 to model wrongful convictions in court; the Gaussian distribution was discovered independently by de Moivre (1733), Gauss (1809), and Laplace (1812).
- Feller [1968, Chapter 4] provides a deep treatment of generating functions for discrete distributions, including elegant proofs of the Poisson limit theorem and moment calculations.

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